IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

UE AML

ADVANCED MACHINE LEARNING

COURSE OBJECTIVE

- Extension of the course « FML : Fundamentals of Machine Learning »
- Theoretical understanding vs. practical application
- Preparing for your future careers in data science
- Orient and enrich your career path
- Develop practical skills through participation in recognized online competitions

6 January	**Online/Kaggle competition**
7 January	**Course "Pattern mining"** **+ free slot for Online/Kaggle competition**
8 January	**Online/Kaggle competition**
9, 12 January	**Course "Meta-Learning"**
13 January	**Online/Kaggle competition**
14 January	**Course "eXplainable AI"** **+ free slot for Online/Kaggle competition**
16, 20 January	**Online/Kaggle competition**
21 January	**Defense: Online/Kaggle competition**

Teachers: Lina FAHED, Romain BILLOT, Sorin MOGA, Matthieu DELAHAYE, Hichem FARAOUN

EVALUATION & COMETENCES

Evaluation methods

- Participate in an online/Kaggle competition + submit/present your work

Evaluated competences

• [BC-02] Develop explainable AI solutions and communicate their insights effectively

• [BC-03] Critically evaluate and select appropriate advanced machine learning techniques for specific problem domains

• [BC-04] Participate competitively in recognized data science challenges and competitions

• [BC-07] Implement and evaluate advanced machine learning techniques on complex, real-world problems

ONLINE / KAGGLE COMPETITION
MACHINE LEARNING / DATA SCIENCE COMPETITION

- Groups of **5 students**
- Look for a challenging competition to participate in
- Discuss/Validate with Profs your choice ==> choice should be validated on **8 January**
- When you ask us to validate the competition on which you would like to participate, ou should :
  - justify why this competition is **challenging**
  - justify it to your future recruiter, and to yours Profs

What is a challenging competition? How to choose a ''good'' competition?

– More **complex / heterogeneous data** compared to what you're used to handle

– **New type of data** that you have not worked on before ==> opportunity to familiarize yourself with and acquire new skills related to data processing

– **New type of modeling/techniques** that you have not practiced before ==> opportunity to acquire new skills related to these models

– Complex, **application-specific modeling**

– .....

Think BIG.. Be ambitious !

EVALUATION MODALITY

Defense: on **21th January**

Duration: you will be given 20mn to present your work, followed by questions

Submissions:

- pdf report
- ppt/pdf presentation
- **GIT** repository

Please show us:

- The entire process you developed/followed to build the **machine learning pipeline**: data analysis, feature engineering, models choice/tuning/evaluation….
- The **challenge related to your competition**: data complexity, new type of data that you have not worked on before, new models that you learned yourselves, the difficulty of the modeling task related to the application…. ==> argument and show us why it is challenging!
- your **ability to step back, to synthesize**….
- Your **ranking or score** among the other Kaggle teams will not be essential in our evaluation. However, it is an important thing that you can highlight in your CV.

RESOURCES

Competition Timeline, you can choose:

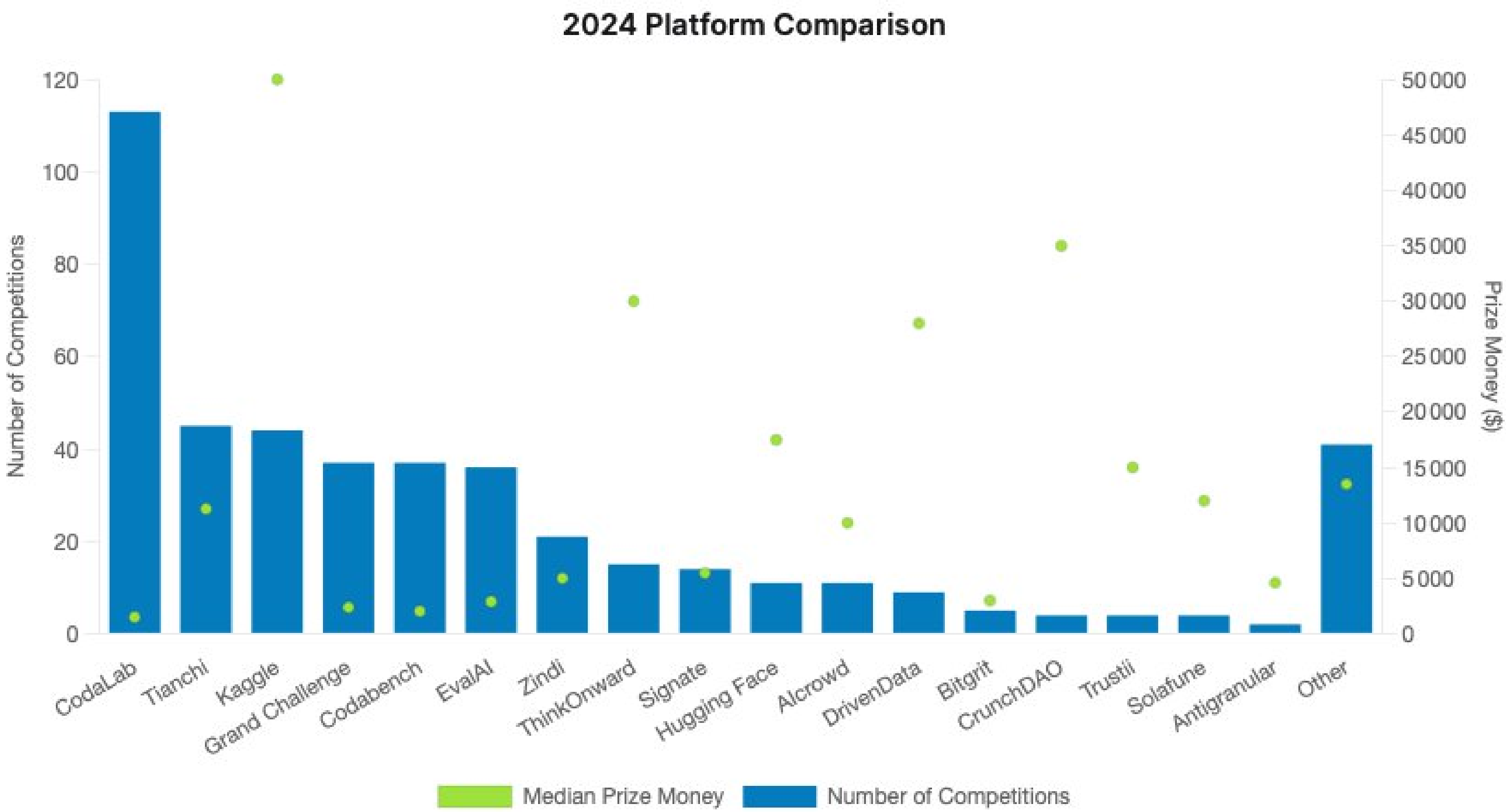
- An ongoing competition with near closing deadline (before the end of the course)
- Or a finished competition with possibility to submit your solution and to be ranked among other participants..

Some **famous competitions websites** :

- **Kaggle : https://www.kaggle.com/competitions**
- ENS : https://challengedata.ens.fr/challenges/challenges_search
- ...

Some resources to find **complementary datasets**

- https://www.archieinitiative.org/
- https://www.data.gouv.fr/fr/
- https://huggingface.co/datasets
- ....



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https://mlcontests.com/state-of-machine-learning-competitions-2024/